Matching the Target Is Not Enough: An Ordering Reversal in Gradient-Based Targeted Instruction Selection

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Abstract

Targeted instruction selection uses query-derived signals to identify training examples useful for a target task, but stronger alignment to such signals need not imply greater downstream utility. We introduce Directional Second-Moment Coreset (DSMC), a greedy selector whose achieved target-gradient discrepancy can be measured directly. On a controlled MMLU stress test, DSMC's advantage over round-robin baselines appears on Llama-2-7B but does not transfer to Llama-3.2-3B. The central result is a stronger BBH counterexample: across three query/selection draws on each of two model stacks, DSMC attains the lowest observed target discrepancy among the compared subsets yet underperforms Random in every draw (-2.94 points on Llama-2-7B; -0.89 on Llama-3.2-3B). Evaluation-matched rescoring and a 256-example proxy set disjoint from every selection query both preserve this DSMC–Random discrepancy ordering. On the same query items used to define the targeting signal, post-SFT teacher-forced loss also decreases without corresponding gains in chain-of-thought exact match; a signed first-moment selector exhibits the same qualitative alignment–utility reversal. Thus, for the studied operational first- and second-moment gradient discrepancies, stronger target alignment is not sufficient for better downstream utility.

1 Introduction

Instruction tuning (Wei et al., 2022) draws from heterogeneous pools, whereas deployment may require a narrow capability. Targeted selection uses a small query set to score candidate data through distribution alignment (Liu et al., 2024b), gradient similarity or influence (Xia et al., 2024; Wang et al., 2025), target subspaces (Min et al., 2026), or task-metric signals (Wang et al., 2025; Wu et al., 2025). We ask a narrower question than which representation or selector is predictive on average (Nayak et al., 2026): when one selected set attains a better measurable target discrepancy, does that ordering imply greater utility? For a fixed pool, budget, and training procedure A, this requires

$$D(S_1, Q) < D(S_2, Q) \quad \Longrightarrow \quad U(A(S_1)) \geq U(A(S_2)). \tag{1}$$

Prior methods need not state Equation 1 as a theorem, but some predictive relationship is necessary to interpret better matching as evidence of greater data utility.

Merely observing that a selector loses is insufficient: it may have failed to improve its own proxy. We therefore construct a measurable instrument. For unit-normalized projected gradients u, represent a set by

$$M_P = \mathbb{E}_{u \sim P}[uu^\top]. \tag{2}$$

Under the quadratic kernel $k_2(u, v) = (u^\top v)^2$, empirical maximum mean discrepancy (MMD) is exactly the squared Frobenius distance between directional second moments (Gretton et al., 2012; Muandet et al., 2017). Directional Second-Moment Coreset (DSMC) greedily reduces this discrepancy using its exact one-step marginal, letting us verify target-geometry movement independently of utility.

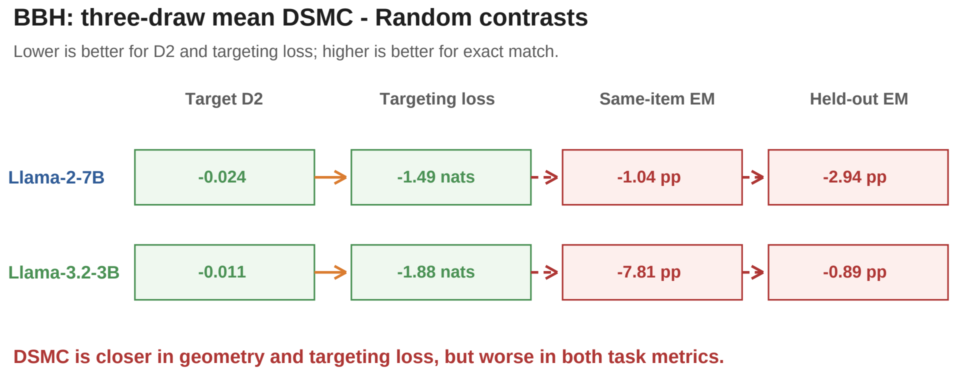

Figure 1: The proxy chain breaks on BBH in both model stacks. DSMC has lower target discrepancy D_2 than Random, and the post-SFT teacher-forced loss underlying the targeting signal decreases, while same-item and held-out exact match fall. Every entry is the three-draw mean DSMC–Random contrast; lower values are favourable for D_2 and targeting loss but unfavourable for exact match. Arrows summarize measured relationships, not an identified causal mechanism.

Method gains are stack-dependent, while the central failure mode is not. On Llama-2-7B MMLU at the 5% budget, DSMC outperforms First-RR and Second-RR by 1.55 and 0.88 points, with wins in all ten target-direction cells. Under a protocol frozen before the second-model computations, these differences reverse on Llama-3.2-3B (-0.18 and -0.31 points over five paired blocks); DSMC and Random are nearly tied on both stacks. We therefore treat DSMC as an instrument rather than a generally superior selector.

In contrast, the BBH geometry–utility reversal transfers across both tested model stacks. Across three query/selection draws, DSMC attains the lowest operational second-moment discrepancy yet loses to Random in every draw on each stack. Llama-2-7B shows a -2.94 percentage-point DSMC–Random gap and Llama-3.2-3B a smaller -0.89 point gap. All BBH SFT means are below their corresponding no-SFT references, so BBH tests ordering among equally trained subsets rather than whether SFT beats the base model. The discrepancy ordering also survives on a 256-example proxy set disjoint from every selection query.

We further separate targeting loss, selection discrepancy, and downstream exact match. On the same 64 query items used to define the signal, post-SFT teacher-forced loss decreases while chain-of-thought exact match does not, ruling out a purely query-to-test account. Because bare-serialization loss behaves differently across stacks, we make no generic claim about cross-entropy.

Contributions.

- We introduce DSMC, an exact-marginal greedy coreset construction with a directly measurable directional second-moment target discrepancy.
- MMLU shows a DSMC advantage over round-robin baselines on Llama-2-7B that does not transfer in a pre-specified Llama-3.2-3B check.
- On BBH, DSMC attains the lowest observed D_2 among the compared subsets in every draw yet underperforms Random in every draw on both model stacks. When the frozen subsets are rescored with evaluation-matched or query-disjoint proxy gradients, DSMC remains closer than Random in every draw; same-item, parser, and task diagnostics localize the failure without claiming a unique mechanism.

2 Target Matching with Directional Second Moments

2.1 Targeted instruction selection

Let $\mathcal{C} = \{x_i\}_{i=1}^{N}$ be a candidate instruction pool and $\mathcal{Q} = \{q_j\}_{j=1}^{M}$ a small target query set. Given a budget K, the goal is to select $\mathcal{S} \subset \mathcal{C}$, $|\mathcal{S}| = K$, such that supervised fine-tuning on $\mathcal{S}$ improves held-out target performance.

Following gradient-based targeted selection (Xia et al., 2024; Nayak et al., 2026; Mirzasoleiman et al., 2020; Killamsetty et al., 2021), we compute per-example gradients at a shared warm-up checkpoint and project them to a reusable low-dimensional datastore. Following the optimizer-aware LESS protocol, candidates and queries use role-specific transforms before entering a common projected directional space:

$$u_{\mathcal{C}}(x) = \frac{\Pi A_{\text{Adam}} g(x)}{\|\Pi A_{\text{Adam}} g(x)\|_2}, \qquad u_{\mathcal{Q}}(q) = \frac{\Pi g(q)}{\|\Pi g(q)\|_2}. \tag{3}$$

Here Π is a shared Rademacher projection and A_{Adam} denotes the candidate-side optimizer-aware transform. The MMD identity below is exact for these resulting unit vectors in the common projected space. The construction should not, however, be interpreted as applying one symmetric raw-gradient representation map to candidates and queries.

2.2 Why a directional second moment?

First-order mean-gradient matching represents a target by $\mathbb{E}[u]$. This signed mean can cancel when useful examples occupy opposite directions, even if they share the same underlying orientation. We instead use the non-centered directional second moment

$$M_P = \mathbb{E}_{u \sim P}[uu^\top]. \tag{4}$$

Because $uu^\top = (-u)(-u)^\top$, this representation captures orientation rather than sign. We deliberately call it a directional second moment, not a covariance: the vectors are normalized and the mean is not subtracted.

Consider the positive semidefinite kernel

$$k_2(u, v) = (u^\top v)^2. \tag{5}$$

Since $\langle uu^\top, vv^\top \rangle_F = (u^\top v)^2$, the biased empirical MMD between a selected set $\mathcal{S}$ and query set $\mathcal{Q}$ is exactly (Gretton et al., 2012; Muandet et al., 2017)

$$\begin{aligned} \text{MMD}^2_{k_2}(\mathcal{S}, \mathcal{Q}) &= \left\| \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} u_s u_s^\top - \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} u_q u_q^\top \right\|_F^2 \\ &\equiv D_2(\mathcal{S}, \mathcal{Q}). \end{aligned} \tag{6}$$

The identity is algebraic: DSMC matches this empirical second moment exactly as defined, but the kernel is not characteristic and does not identify the full gradient distribution (Sriperumbudur et al., 2011; Muandet et al., 2017).

2.3 Exact greedy coreset construction

For a candidate x, define target relevance and accumulated selected-set similarity as

$$r_{\mathcal{Q}}(x) = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} k_2(u_x, u_q), \quad r_{\mathcal{S}}(x) = \sum_{s \in \mathcal{S}} k_2(u_x, u_s). \tag{7}$$

At step $m = |\mathcal{S}|$, the exact one-step marginal for minimizing the biased MMD selects

$$x^\star = \arg\max_{x \notin \mathcal{S}} \left[r_{\mathcal{Q}}(x) - \frac{r_{\mathcal{S}}(x) + k_2(u_x, u_x)/2}{m+1} \right]. \tag{8}$$

For unit vectors $k_2(u_x, u_x) = 1$. The first term rewards query alignment and the second penalizes selected-set self-similarity. We use the deterministic exact marginal throughout; we make no stochastic-greedy or submodularity guarantee. The construction is related to kernel herding and greedy kernel-mean approximation (Chen et al., 2010; Bach et al., 2012), but Equation 8 is the exact marginal of the finite-sample objective used here.

Why proxy ordering need not survive training. For one small SGD step on the same query loss, $L_Q(\theta - \eta g_S) - L_Q(\theta) \approx -\eta g_Q^\top g_S$ motivates alignment locally. Our pipeline additionally applies role-specific transforms, greedy construction, multi-step Adam in a LoRA subspace, generation, and exact match. Equation 1 would require this chain to preserve subset order; our results do not contradict the local Taylor approximation.

Role in this paper. DSMC is a measurement instrument, not a claim of general selector superiority. If it lowers D_2 relative to compared subsets without improving utility, the reversal cannot be dismissed as failure to move its own proxy.

3 Experimental Design

3.1 Candidate pool, features, and training

The candidate pool is the processed pool released with LESS (Xia et al., 2024): 270,679 examples from Flan v2 (Longpre et al., 2023) (100,000), its CoT mixture (100,000), Dolly (Databricks, 2023) (15,011), and OASST1 (Köpf et al., 2023) (55,668). For each model stack, a LoRA (Hu et al., 2022) warm-up checkpoint is used to construct a candidate-gradient datastore. We use an 8,192-dimensional Rademacher projection with seed 123, unit-normalize the projected vectors, and generate query-gradient caches separately for each draw. The selected indices are materialized into static subsets; all downstream comparisons restart from the corresponding base model rather than continuing the warm-up run. We study Llama-2-7B and Llama-3.2-3B model stacks (Touvron et al., 2023; Grattafiori et al., 2024; Meta AI, 2024), implemented through LLaMA-Factory (Zheng et al., 2024).

Downstream SFT uses LoRA rank 128, alpha 512, dropout 0.05 on q/k/v/o_proj, learning rate 2×10^{-5}, linear decay, 3% warm-up, bfloat16, effective batch size 128, and four epochs. The 5% budget is $K = 13{,}533$ (420 optimizer steps) and the 1% budget is $K = 2{,}707$ (84 steps). Full configurations, prompt templates, hashes, and data-processing details are provided in Appendix J.

3.2 Target families

MMLU controlled stress test. We construct ten globally disjoint 64-example query sets: five STEM-majority draws (51 STEM, 13 humanities) and five humanities-majority draws with the proportions reversed. Evaluation is balanced accuracy

$$\mathrm{BalAcc} = \tfrac{1}{2}(\mathrm{Acc}_{\mathrm{STEM}} + \mathrm{Acc}_{\mathrm{Humanities}})$$

over held-out MMLU subjects (Hendrycks et al., 2021). This is our equal-category aggregate, not the benchmark's standard overall accuracy. Draw index is paired with SFT seed $\{42, 1, 2, 3, 4\}$; for comparisons shared across the two skew directions, the primary descriptive unit is the five direction-averaged draw-index blocks.

BBH external validation. Using the BIG-Bench Hard suite (Suzgun et al., 2023), we construct within each task family a disjoint query-reservoir/evaluation partition of the official examples. The query reservoir contains 1,302 examples and the frozen held-out partition contains 5,209. The evaluation harness exposes 27 subtasks spanning 23 conceptual BBH task families, because two families contain three size-specific variants. Three stratified 64-example query draws define the selection signals; we report the size-weighted micro exact-match aggregate over the custom held-out partition using a pinned chain-of-thought few-shot protocol. Each query/selection draw is crossed with two SFT seeds, which are averaged within the draw. The statistical unit is therefore the draw ($n = 3$), not the six seed-level runs.

3.3 Comparators and reporting policy

The controlled gradient baselines are: first-order mean-gradient TopK (LESS-style (Xia et al., 2024)), first- and second-order relevance TopK, first- and second-order round-robin

Table 1: Llama-2 MMLU representation × selector attribution. Balanced accuracy at one fixed seed for each skew direction. In this stack, the second-order representation improves both TopK and MMD; the smaller DSMC–Second-TopK increment is within the measured training-noise scale and is not treated as a stand-alone statistical claim.

Target	1st-TopK	1st-MMD	2nd-TopK	DSMC
STEM-majority	39.77	38.47	40.68	41.10
Humanities-majority	39.31	38.24	39.92	40.54

(RR) (Nayak et al., 2026), GIST whitening on the shared projected datastore (GIST-SharedProj (Min et al., 2026)), and a task-metric adaptation of NICE (NICE-MMLU-EM (Wang et al., 2025)). Because several are shared-protocol adaptations rather than full official end-to-end reproductions, we use these qualified names in the paper. Target-independent controls are uniform Random-K, length-matched Random on MMLU, a sequence/label-length-matched Random control on BBH, and a shared no-SFT reference.

The study is descriptive. We report draw-level means, paired differences, and win counts; we do not report significance tests over $n = 3$ or $n = 5$ blocks. Subtask breakdowns and individual SFT seeds are diagnostics, not independent replicates.

4 Method Gains Are Stack-Dependent on MMLU

4.1 Llama-2 positive control

We first evaluate whether second-moment matching provides a meaningful targeted-selection signal on MMLU. Table 1 factors the method into target representation (signed first order versus directional second order) and selection rule (relevance TopK versus MMD coreset). Moving to the second-order representation improves both selectors in both skew directions: +0.61–0.91 points under TopK and +2.30–2.63 points under MMD. The selector interaction is representation dependent: first-order MMD hurts relative to first-order TopK, while second-order MMD adds 0.42–0.62 points over second-order TopK. The latter comparison is single-seed and small, so this is a Llama-2 positive control, not a cross-stack method claim.

Across ten target draws at the 5% budget, DSMC beats LESS-style TopK and both RR baselines in every draw, and beats the examined GIST and NICE adaptations in nine of ten draws (Appendix B). Relative to Second-RR, the mean gain is 0.88 points and holds in all ten cells, supporting a complementary coreset effect at this budget. The positive claim is deliberately scoped: in the studied Llama-2 MMLU setting, directional second moments materially improve target-aware selection.

The absolute comparison creates the first reversal. Against Random-K, DSMC has a -0.01 point mean difference at 5%, winning two of five direction-averaged blocks; length-matched Random gives the same qualitative conclusion. Tightening the budget does not rescue target awareness. At 1%, the DSMC–Random gap becomes -0.77 points and no query-direction cell favours DSMC. The extra MMD gain over Second-RR also contracts to 0.17 points and five of ten cells.

Most targeted methods fall below no-SFT at both budgets. However, the 1% arm is a positive-transfer ordering reversal in observed means: DSMC (40.17) and Random (40.95) both exceed no-SFT (40.03), while DSMC is closer to the leave-one-draw-out balanced target geometry in all ten draws. Thus the ordering failure is not restricted to harmful SFT, although this deliberately skewed stress test complements rather than replaces same-family BBH validation. An equal-step sensitivity repeats the 1% subsets to 420 steps; both DSMC and Random fall to approximately 38.3%, so extra optimization does not rescue the targeted subset (Appendix F).

Table 2: Cross-stack MMLU comparison at the 5% budget. DSMC's advantage over the two RR baselines is consistent on Llama-2-7B but does not transfer to Llama-3.2-3B. DSMC and Random have nearly tied observed means on both stacks. Win counts use the same five direction-averaged draw-index blocks for both stacks; no significance test is claimed.

Stack	DSMC − First-RR	DSMC − Second-RR	DSMC − Random
Llama-2-7B	+1.55 (5/5)	+0.88 (5/5)	−0.01 (2/5)
Llama-3.2-3B	−0.18 (2/5)	−0.31 (1/5)	+0.12 (3/5)

Table 3: Held-out BBH exact match. Means first average the two SFT seeds within each of three query/selection draws. All method means are below the corresponding no-SFT reference. DSMC loses to Random in every draw on both model stacks, while achieving the lowest target second-moment discrepancy among the compared subsets in every draw.

Stack	Method	Draw 0	Draw 1	Draw 2	Mean	Δ vs. no-SFT
Llama-2-7B	No-SFT		39.64			–
	Random-K	39.30	38.78	39.67	39.25	−0.39
	First-RR	36.24	37.61	37.61	37.15	−2.49
	Second-RR	36.80	37.00	36.80	36.87	−2.77
	DSMC	36.36	36.47	36.10	36.31	−3.33
Llama-3.2-3B	No-SFT		47.11			–
	Random-K	47.32	46.61	47.17	47.03	−0.08
	First-RR	46.11	46.52	47.04	46.56	−0.55
	Second-RR	46.62	46.23	47.18	46.68	−0.44
	DSMC	45.98	45.88	46.57	46.14	−0.97

4.2 Pre-specified cross-stack check

The cross-stack check does not support a transferable DSMC advantage. Mean balanced accuracy is 47.95 for Second-RR, 47.82 for First-RR, 47.64 for DSMC, and 47.52 for Random, while the no-SFT reference is 49.47. All four methods lie within 0.44 points and below base. Table 2 shows the sharper contrast: DSMC's Llama-2 advantage over both RR arms reverses on Llama-3.2, whereas its difference from Random remains close to zero on both stacks. Thus the method-level advantage of DSMC does not transfer. In contrast, the BBH geometry–utility counterexample below does.

5 External Reversal on BBH

MMLU queries are deliberately skewed, leaving open the possibility that the selector faithfully follows an unrepresentative query sample. BBH removes this specific explanation: queries and evaluation examples are held-out samples from the same task family. In this setting, the geometry–utility reversal persists (Table 3). On Llama-2-7B, Random reaches 39.25% micro exact match, while DSMC reaches 36.31%, a −2.94 point paired gap with zero of three draws favouring DSMC. Both RR variants also outperform DSMC; together with Table 2, this shows that a favourable DSMC ranking is neither cross-task nor cross-stack.

The direction repeats on Llama-3.2-3B. Random remains best at 47.03%, and DSMC is last among the four shared arms at 46.14%. The −0.89 point gap is negative in all three draws. The magnitude is substantially attenuated: all methods are within one point of the no-SFT model, so the two stacks should not be presented as equally dramatic. They nevertheless agree on the direction of the DSMC–Random gap.

All SFT means are below no-SFT. On Llama-2, Random is only 0.39 points below the shared base reference, while target-aware methods are 2.49–3.63 points below it. On Llama-3.2, Random is 0.08 points below base and DSMC 0.97 below. These are shared reference lines rather than replicated treatments, so we describe the observed means rather than claim inferentially established negative transfer. The SFT infrastructure is not uniformly detrimental: under the same Llama-2 pipeline, DSMC and Random are slightly above the no-SFT reference on MMLU at 5%, and Random is 0.92 points above it at 1%. We therefore treat BBH and Llama-3.2 MMLU as observed negative-transfer regimes, not evidence that the training implementation cannot produce gains.

Matched-format control. On Llama-2, constraining Random to match DSMC's coarse sequence-length × loss-bearing-label-length histogram lowers Random from 39.25% to 38.05%, but still leaves it 1.74 points above DSMC. This is consistent with instruction-format or correlated provenance composition contributing to the gap. It is not a causal decomposition: the control changes several correlated properties simultaneously. Full results are in Appendix F.

6 Where the Surrogate Chain Breaks

6.1 DSMC attains the lowest observed target discrepancy

For every BBH draw, we compute the operational discrepancy used by DSMC,

$$D_2(\mathcal{S}, \mathcal{Q}_d) = \|M_{\mathcal{S}} - M_{\mathcal{Q}_d}\|_F^2. \tag{9}$$

On Llama-2, mean D_2 is 0.0863 for DSMC and 0.1098 for Random; on Llama-3.2 it is 0.0644 and 0.0752. More importantly, DSMC has the lowest D_2 among the compared subsets in all three draws of both stacks, while being worse than Random downstream in all three draws. This direct 3/3-versus-3/3 counterexample is the primary geometry result; descriptive rank correlations are reported only in Appendix C.

This shows that the downstream reversal is not explained merely by DSMC failing to reduce its intended discrepancy relative to the compared subsets; it does not imply global optimality of the greedy solution. In MMLU, DSMC is also closer than Random to a leave-one-draw-out balanced reference in all ten draws at both budgets (Appendix C). Thus the observed failure is not simply that DSMC follows a skewed query away from the balanced reference.

Query-disjoint proxy geometry. We post-hoc score frozen subsets against 256 examples sampled after excluding all selection queries, under a protocol frozen before proxy-gradient extraction. Without reselection or SFT, DSMC remains closer than Random under D_2, and First-RR under signed D_1, in all three draws on both stacks; both remain worse downstream. The reversal therefore extends beyond selection-example geometry (Appendix D).

Sign-sensitive first-moment diagnostic. The quadratic kernel is invariant to $u \mapsto -u$, so we also examine the signed first-moment discrepancy

$$D_1(\mathcal{S}, \mathcal{Q}) = \|\mathbb{E}_{\mathcal{S}}[u] - \mathbb{E}_{\mathcal{Q}}[u]\|_2\,. \tag{10}$$

First-RR is closer than Random under D_1 in all three draws on both stacks, yet its held-out BBH mean is lower than Random by 2.10 points on Llama-2 and 0.48 points on Llama-3.2. The observed alignment–utility reversal is therefore not confined to the sign-invariant second moment. This diagnostic does not establish failure for every sign-sensitive representation; full draw-level values are in Appendix C; the mean gaps (0.471 vs. 0.591 and 0.459 vs. 0.498) are not epsilon effects.

Evaluation-matched target serialization. To test whether the DSMC minimum is induced by the chat serialization used for target extraction, we re-extract only target gradients under the bare evaluation serialization and rescore frozen subsets. Despite low cross-context row cosine (0.25–0.36), DSMC retains the lowest D_2 in all draws on both stacks while losing to Random downstream. The full Llama-3.2 ranking changes in every draw, so we claim robustness only of the DSMC–Random counterexample, not serialization-invariant rankings (Appendix E).

6.2 The query loss underlying the targeting signal also improves

Gradient-based targeting uses a differentiable final-answer objective to construct query gradients. We therefore recompute teacher-forced final-answer cross-entropy on the same query items, under the exact wrapped serialization used by each stack's targeting pipeline. On Llama-2, all four target-aware methods reduce this loss by 0.68–1.40 nats, while Random increases it by 0.33. On Llama-3.2, the target-aware methods reduce it by 3.19–3.40 nats and Random by only 1.37. The second stack therefore gives a magnitude split rather than a sign split; on both stacks, the targeted methods produce larger reductions in the query loss from which the targeting gradients are derived.

6.3 Task utility does not follow on the same examples

Comparing query loss to held-out exact match changes both metric and examples. We consequently evaluate chain-of-thought exact match on the same 64 query items. For Llama-2, every target-aware method's mean query exact match falls relative to base (-3.65 to -6.51 points) despite the substantial cross-entropy reductions. On Llama-3.2, all three targeted methods again have negative mean changes, with DSMC falling by 6.51 points; Random changes by $+1.30$ points. The individual 64-item draw metrics are coarse—one item is 1.56 points—but the mean separation is consistent with the held-out ranking.

The same-item analysis rules out a purely query-to-test generalization account: the downstream metric does not improve even on the examples that form the targeting signal. It does not rule out every form of overfitting, nor does it identify the missing mechanism.

Task-level breadth. The draw remains the primary experimental unit. A secondary 27-subtask breakdown shows that DSMC trails Random on 23 subtasks and leads on four for Llama-2 (macro difference -2.89 points). On Llama-3.2 it trails on 17 and leads on ten (macro difference -1.10). A descriptive bootstrap over subtasks gives intervals of $[-4.68, -1.41]$ and $[-2.57, +0.27]$ points, respectively. Thus the Llama-2 gap is broad across subtasks, whereas the smaller Llama-3.2 effect is more heterogeneous. Regrouping size variants into the 23 conceptual BBH families gives the same reading: 20/23 and 14/23 families favour Random.

Parser-validity audit. A post-hoc audit of all frozen generations finds invalid-rate contrasts of $+0.03$ points on Llama-2 and -0.78 on Llama-3.2. DSMC remains below Random conditional on standard-valid outputs (-3.13, -1.36 points) and under frozen conservative recovery (-3.08, -0.84), so parsing does not explain the gap (Appendix G).

6.4 Serialization narrows the loss-based diagnostic

The targeting loss uses each stack's LLaMA-Factory wrapper (Zheng et al., 2024), whereas BBH generation uses a pinned bare evaluation context implemented through the language-model evaluation harness (Biderman et al., 2024). Recomputing final-answer cross-entropy on that bare context yields different behavior across stacks. On Llama-2, no method improves bare-context CE, although targeted methods degrade it much less than Random. On Llama-3.2, target-aware methods do improve bare-context CE while Random does not. Therefore we do not claim that final-answer cross-entropy is inherently misaligned with chain-of-thought generation. The supported claim is narrower: the targeting loss used to construct query gradients can decrease on the defining examples without a corresponding improvement in chain-of-thought exact match, and this relationship is serialization- and model-stack-dependent.

7 Related Work

Instruction data selection. General-purpose instruction selection commonly scores data for quality, difficulty, and diversity. DEITA jointly studies complexity, quality, and diversity for data-efficient alignment (Liu et al., 2024a). Get More for Less studies selection for the warm-up stage before target fine-tuning (Kang et al., 2024). TSDS uses a small target set to define an optimal-transport distribution-alignment objective (Liu et al., 2024b). Our focus is on testing whether attaining a better query-derived proxy value is sufficient for utility.

Distribution matching and gradient coresets. MMD and kernel mean embeddings define set-level distribution discrepancies (Gretton et al., 2012; Muandet et al., 2017); characteristic-kernel results specify when these discrepancies identify distributions (Sriperumbudur et al., 2011). Kernel herding greedily approximates a target kernel mean (Chen et al., 2010; Bach et al., 2012), while CRAIG and GRAD-MATCH use gradient geometry for data-efficient training (Mirzasoleiman et al., 2020; Killamsetty et al., 2021). Our novelty is not the quadratic-kernel identity or greedy moment matching in isolation, but their use as a directly measurable intervention for testing whether stronger target-gradient alignment is sufficient for downstream utility.

Targeted and influence-based selection. LESS builds reusable low-dimensional gradient datastores and uses optimizer-aware gradient similarity for targeted instruction tuning (Xia et al., 2024). NICE replaces next-token-loss influence with a policy-gradient construction tied to non-differentiable evaluation metrics (Wang et al., 2025). GIST recovers a target-gradient subspace and scores candidate alignment under coupled optimization geometry (Min et al., 2026). GIO uses examples representing a target distribution to optimize a scalable gradient-information objective for dataset selection (Everaert & Potts, 2024). A recent controlled study separates representations from selection algorithms and finds that gradient representations are often predictive, but no single selector dominates across settings (Nayak et al., 2026). Where that study asks which representations and selectors are predictive across settings, we ask whether attaining a better value of a measurable target geometry is itself sufficient for greater downstream utility.

Surrogate mismatch and strong Random baselines. ROSE motivates reward-oriented selection by documenting non-monotonicity between instruction-tuning loss and task reward (Wu et al., 2025); NICE makes a related move from differentiable training loss to task metrics (Wang et al., 2025). Controlled studies also find that sophisticated selectors may fail to consistently outperform Random (Diddee & Ippolito, 2025; Xia et al., 2025). Together, our diagnostics establish an empirical counterexample in which lower target discrepancy and lower targeting loss do not yield higher task utility.

8 Limitations

The study covers two model stacks, one candidate pool, and two target families; a stack changes model, tokenizer, and serialization together. BBH has only three query/selection draws and the 3B effect is small. On Llama-3.2 MMLU, all methods are within 0.44 points and below no-SFT. MMLU also couples draw index to SFT seed, so five direction-averaged blocks, not ten cells, carry shared comparisons.

Candidates use Adam-aware features whereas queries use raw SGD gradients, so claims concern this operational push-forward geometry rather than symmetric raw-gradient matching. Evaluation-matched rescoring does not remove this asymmetry, and the finer Llama-3.2 ranking changes in every draw.

Several baselines are controlled adaptations rather than full end-to-end reproductions. Contamination checks are lexical, not semantic. The diagnostics do not identify a unique causal mechanism: format and provenance remain plausible, the query-disjoint proxy only rescores frozen subsets from the query reservoir, and the parser audit tests extraction rather than semantic correctness.

9 Conclusion

What transfers across the two tested model stacks is not DSMC's method advantage, but a counterexample to the sufficiency of the studied selection objective. Across the positive-transfer MMLU 1% comparison and the negative-transfer BBH regime, lower values of the studied operational first- and second-moment gradient discrepancies do not guarantee higher downstream utility. A better value of a query-derived selection objective should therefore be treated as progress only after its relationship to the downstream task metric has been independently validated.

AI use statement

Generative AI tools were used to summarize and cross-check existing literature, inspect repository artifacts, suggest paper structure and wording, draft and edit parts of the manuscript, and assist with software implementation and analysis scripts. Earlier in the research workflow, generative AI tools also provided feedback on experimental design and result interpretation. The authors reviewed all AI-assisted text, code, and claims against the underlying implementations, configurations, manifests, and result files; all reported numerical results were computed from experiment artifacts rather than generated by an AI system. The authors take responsibility for the final content, including all AI-assisted material.

Ethics statement

This work studies data selection for publicly available language-model benchmarks and does not involve human-subject data collection. The main risks are those shared by instruction tuning, including inherited dataset biases, benchmark contamination, and overgeneralization from a limited experimental scope. We report negative results, adapted-baseline scope, and contamination audit limits explicitly to reduce the risk of overstating practical conclusions.

Reproducibility statement

Appendix J documents the data composition, target-draw construction, feature protocol, selection rules, training hyperparameters, evaluation pins, statistical units, and integrity checks. The supplementary material will include anonymized code, exact run configurations, frozen manifests and hashes, paper tables generated from committed result files, and the timestamped, pre-outcome protocol specifications for the BBH and second-model experiments.

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A Method Derivations

A.1 MMD identity

Let $\widehat{M}_{\mathcal{S}} = |\mathcal{S}|^{-1} \sum_{s\in\mathcal{S}} u_s u_s^\top$ and define $\widehat{M}_{\mathcal{Q}}$ analogously. Expanding the squared Frobenius norm gives

$$\begin{aligned}
\|\widehat{M}_{\mathcal{S}} - \widehat{M}_{\mathcal{Q}}\|_F^2 = \frac{1}{|\mathcal{S}|^2} \sum_{s,s'} \langle u_s u_s^\top, u_{s'} u_{s'}^\top \rangle_F \\
+ \frac{1}{|\mathcal{Q}|^2} \sum_{q,q'} \langle u_q u_q^\top, u_{q'} u_{q'}^\top \rangle_F \\
- \frac{2}{|\mathcal{S}||\mathcal{Q}|} \sum_{s,q} \langle u_s u_s^\top, u_q u_q^\top \rangle_F \qquad (11)
\end{aligned}$$

$$= \frac{1}{|\mathcal{S}|^2} \sum_{s,s'} (u_s^\top u_{s'})^2 + \frac{1}{|\mathcal{Q}|^2} \sum_{q,q'} (u_q^\top u_{q'})^2 - \frac{2}{|\mathcal{S}||\mathcal{Q}|} \sum_{s,q} (u_s^\top u_q)^2. \qquad (12)$$

This is the biased empirical MMD under k_2.

A.2 Greedy marginal

Terms depending only on $\mathcal{Q}$ can be dropped. Comparing the objective after adding x to a set of size m yields Equation 8 after positive rescaling. The implementation maintains $r_{\mathcal{S}}(x)$ incrementally and masks selected or invalid rows. Zero and non-finite feature rows are rejected before selection.

B Additional MMLU Results

Table 4: Llama-2 MMLU target-draw results. Mean equal-domain balanced accuracy over ten query draws (five paired draw-index blocks). The no-SFT reference is shared, not a replicate.

Method	5%	1%
No-SFT	40.03	40.03
LESS-style TopK	39.41	39.37
First-RR	38.80	38.20
Second-RR	39.48	40.00
GIST-SharedProj	38.85	39.45
NICE-MMLU-EM	39.25	39.42
DSMC	40.36	40.17
Random-K	40.37	40.95
Length-matched Random	40.42	40.80

Pre-specified Llama-3.2 outcome. The 5% cross-stack arm completed 35/35 adapters and 36/36 evaluations with no failures. The statistical unit is five direction-averaged draw-index blocks. The protocol was committed before the second-model computations. The no-SFT balanced reference is 49.47; mean balanced accuracy is 47.95 for Second-RR, 47.82 for First-RR, 47.64 for DSMC, and 47.52 for Random. DSMC minus First-RR is -0.18 points (2/5 blocks), minus Second-RR is -0.31 (1/5), and minus Random is $+0.12$ (3/5). This is the pre-specified outcome in which the Llama-2 method advantage does not transfer.

Lower budget and equal-step sensitivity. On Llama-2, DSMC minus Random changes from -0.01 points at 5% to -0.77 at 1%, with 0/10 1% cells favouring DSMC. Repeating the 1% DSMC and Random subsets to the 420 steps used by the 5% arm reduces both to approximately 38.3%, about 1.8 points below the shared no-SFT reference. Thus the low-budget comparison is not rescued by simply increasing optimizer steps.

C Additional Geometry Diagnostics

On MMLU, a leave-one-draw-out balanced reference gives mean D_2 values of 0.1538 for DSMC and 0.1762 for Random at 5%, and 0.1522 versus 0.1764 at 1%; DSMC is closer in all ten draws at each budget. On BBH, the descriptive Spearman correlation between D_2 and accuracy is positive in all three Llama-2 draws (0.77, 0.83, 0.89 over six methods) and all three Llama-3.2 draws (0.80, 1.00, 0.20 over four methods). These ranking samples are small and no inferential interpretation is attached to them.

When the size-specific logical-deduction and shuffled-object subtasks are regrouped into the 23 conceptual BBH families, DSMC wins/ties/loses 3/0/20 families on Llama-2 and 9/0/14 on Llama-3.2. Family-macro differences are -2.94 and -1.23 points, with descriptive task-family bootstrap intervals $[-4.94, -1.25]$ and $[-2.91, +0.26]$.

Table 5: Signed first-moment discrepancy on BBH. $D_1 = \|\mathbb{E}_{\mathcal{S}}[u] - \mathbb{E}_{\mathcal{Q}}[u]\|_2$ is sign-sensitive. First-RR is closer than Random in every draw, yet its downstream mean is lower on both stacks.

Stack	Draw	DSMC	First-RR	Random
Llama-2	0	0.4886	0.4763	0.5957
	1	0.4972	0.4861	0.6032
	2	0.4610	0.4516	0.5741
Llama-3.2	0	0.4404	0.4611	0.5012
	1	0.4599	0.4789	0.5179
	2	0.4081	0.4374	0.4736

Table 6: Task-level DSMC–Random heterogeneity. Win/tie/loss counts and macro statistics average SFT seeds and query draws within each of the 27 harness subtasks. Bootstrap intervals resample subtasks and are descriptive; they do not replace the pre-specified draw-level unit.

Stack	W/T/L	Macro Δ	Median Δ	Task-bootstrap 95%
Llama-2	4/0/23	−2.89	−1.67	[−4.68, −1.41]
Llama-3.2	10/0/17	−1.10	−0.92	[−2.57, +0.27]

D Query-Disjoint Proxy Geometry

This post-hoc target-only analysis follows a protocol frozen before the proxy gradients were extracted. Starting from the 1,302-example BBH query reservoir, we exclude the 185 distinct examples appearing in any selection query and sample 256 examples by deterministic proportional stratification over the 27 harness subtasks (seed 20260827). The resulting proxy set is disjoint from every selection query. We rescore only the frozen subsets; no selection, training, or downstream evaluation is rerun.

Table 7: Geometry against the query-disjoint BBH proxy set. Lower is better. The frozen DSMC subset is closer than Random under D_2 in every stack–draw cell; First-RR is closer than Random under signed D_1 in every cell.

Stack	Draw	Metric	DSMC	First-RR	Second-RR	Random
Llama-2	0	D_1	0.46684	0.45508	0.50589	0.57679
		D_2	0.07067	0.07250	0.07482	0.09445
	1	D_1	0.46540	0.45579	0.50108	0.57522
		D_2	0.07069	0.07256	0.07409	0.09360
	2	D_1	0.46524	0.45562	0.50093	0.57786
		D_2	0.07067	0.07237	0.07392	0.09412
Llama-3.2	0	D_1	0.41998	0.44544	0.43849	0.48214
		D_2	0.05035	0.06087	0.05868	0.06056
	1	D_1	0.42084	0.44568	0.43871	0.48190
		D_2	0.05037	0.06083	0.05886	0.06132
	2	D_1	0.41935	0.44450	0.44258	0.48403
		D_2	0.05034	0.06230	0.06010	0.06103

The proxy comes from the held-out query reservoir, not the 5,209-example downstream evaluation partition. These results establish frozen-subset geometry ordering only and do not show what would happen after selecting and training directly against the proxy target.

E Evaluation-Matched Target-Geometry Sensitivity

This post-hoc check changes only target serialization; it does not rerun selection or SFT. Mean row cosine between operational and evaluation-matched target gradients is 0.335/0.342/0.363 for Llama-2 and 0.249/0.270/0.272 for Llama-3.2.

Table 8: Evaluation-matched D_2 for frozen BBH subsets. Lower is better. The frozen DSMC subset is closest in all six stack–draw cells, but the non-DSMC ranking changes in all three Llama-3.2 draws.

Stack	Draw	DSMC	First-RR	Second-RR	Random
Llama-2	0	0.06675	0.06830	0.07102	0.08875
	1	0.08230	0.08393	0.08586	0.10358
	2	0.05655	0.05800	0.05989	0.07831
Llama-3.2	0	0.07985	0.09134	0.08921	0.08847
	1	0.07660	0.08809	0.08619	0.08598
	2	0.07124	0.08414	0.08203	0.08051

F Sensitivity and Falsification Checks

Matched Random controls. MMLU length-matched Random gives the same qualitative target-awareness result as plain Random. On Llama-2 BBH, sequence×label-length-matched Random scores 38.05, between plain Random (39.25) and DSMC (36.31). The control changes correlated provenance and content as well as format, so it does not support a causal percentage decomposition.

Same-item and bare-context diagnostics. On Llama-2, targeted methods reduce the wrapped targeting loss by 0.68–1.40 nats but their same-item CoT exact match changes by -3.65 to -6.51 points. Under the bare evaluation context, no method improves CE; targeted methods degrade it less than Random. On Llama-3.2, targeted methods reduce wrapped targeting loss by 3.19–3.40 nats and also reduce bare CE by 0.41–0.57, while their mean same-item CoT changes remain negative. This is why the loss-based diagnostic is tied to the tested targeting pipeline, not cross-entropy in general.

Task exposure. Per-task query frequency has negative descriptive correlations with targeted-vs-Random subtask deltas (-0.210, -0.280, -0.224 across the three draws). We find no evidence for a simple account in which greater query exposure protects a BBH task. This diagnostic is exploratory and does not identify an alternative mechanism.

G Parser-Validity Audit

We audit every saved BBH generation without regenerating outputs. Conditional EM is computed only over examples accepted by the standard harness parser. For standard-invalid outputs, a frozen conservative rule inspects only the final 500 characters and extracts a type-constrained final candidate: the last parenthesized option, boolean/yes–no token, or standalone number when applicable, otherwise the last explicit answer marker or nonempty line. Unrecovered outputs remain incorrect. This secondary rule is not a replacement for the primary metric.

The DSMC–Random invalid-rate difference is $+0.03$ percentage points on Llama-2 and -0.78 on Llama-3.2. The corresponding conditional-EM differences are -3.13 and -1.36 points, and conservative-EM differences are -3.08 and -0.84. Thus the observed utility ordering is not produced merely by more DSMC outputs failing the standard parser. The audit does not establish semantic correctness beyond these extraction rules.

Table 9: BBH parser-validity audit. Rates first average the two SFT seeds within each draw and then the three draws. Conservative EM applies the frozen recovery rule only to standard-invalid outputs.

Stack	Method	Standard valid	Conditional EM	Conservative EM
Llama-2	DSMC	93.46	38.85	36.42
	Random	93.50	41.98	39.50
Llama-3.2	DSMC	94.42	48.87	46.77
	Random	93.64	50.23	47.61

H Baseline Implementations

LESS-style TopK uses one shared checkpoint and mean signed gradient similarity; it is not the official multi-checkpoint LESS pipeline. GIST-SharedProj applies the paper's target-subspace whitening construction to the experiment's shared normalized projected caches. NICE-MMLU-EM uses eight Monte Carlo samples and an exact-match task reward in the common feature protocol. These adaptations are intended to isolate representation and selection effects under controlled features; the paper does not claim exact end-to-end reproduction of the official systems.

I Contamination and Prompt Audits

Against MMLU and held-out BBH, normalized exact matching finds zero candidate duplicates. Full-pool fuzzy screening finds zero pairs above Jaccard 0.3; seven MMLU and five BBH 13-gram hits were manually reviewed and judged boilerplate false positives. The BBH few-shot audit finds no exact duplicates and is recorded as pass-with-disclosure for fuzzy near-duplicates. Prompt parity, few-shot leakage, and token-truncation audits are included in the supplementary artifact. A semantic embedding nearest-neighbor contamination audit was not performed, so these checks are lexical screens rather than a complete decontamination certificate.

J Reproducibility Details

The candidate pool contains 270,679 examples. Projected gradients use dimension 8,192 and seed 123. MMLU target draws are globally disjoint; BBH uses three stratified 64-example query draws and a 5,209-example held-out set. Frozen manifests record target files, gradient tensors, selections, materialized subsets, adapters, and evaluation-result hashes. All BBH evaluations pass 27-subtask/5,209-example gates, and all Llama-3.2 MMLU evaluations pass 57-subject gates. Selection-level stability was checked after target-gradient non-bit-reproducibility was observed; the worst selector changed 5 of 2,707 indices. A paper-level artifact manifest maps each major figure and table to its committed result files, analysis code, and frozen protocol.

J.1 Compute and selection overhead

The main practical asymmetry is therefore not the greedy selector alone but the model-in-the-loop feature pipeline: Random requires neither warm-up nor gradient datastores. The anonymous supplement will include sanitized configurations and manifests. Before release, absolute cluster paths, Git metadata, user names, and repository links must be removed.

Table 10: Representative measured or launch-accounted costs on eight NVIDIA H20 GPUs. Candidate datastores are reusable across target draws and tasks for a fixed model stack. Times for target gradients and selectors are coarse stage-level accounting rather than microbenchmarks.

Stage	Reuse scope	Representative cost
Warm-up checkpoint	per model stack	$\sim$1.5 GPU-h
Candidate gradient datastore	across targets	$\sim$6–8 GPU-h per stack
Three BBH target caches	per query draws	$\sim$1 GPU-h
DSMC + First/Second-RR	per three draws	$\sim$1.5 GPU-h
Random-K sampling	per subset	negligible
Llama-3.2 BBH SFT, 24 adapters	per selected subsets	3.39 GPU-h measured
Llama-3.2 MMLU SFT, 35 adapters	per selected subsets	16.88 GPU-h measured
Llama-2 BBH SFT, 36 adapters	per selected subsets	13.18 GPU-h measured